

BT101

Introductory Biology

(Lecture 3)

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Introduction



Evolution of life:

☐ **Origin of Life**

- ☐ Darwin's concepts of evolution
- ☐ Biodiversity

Cell, the structural and functional unit of life:

- ☐ Three domains of life
- ☐ Cell types, cell organelles and structure
- ☐ Basic biomolecules of cell

Introduction



- ❑ Organisms are divided into **two** main classes, initially defined by whether they contain a nucleus.
- ❑ Prokaryotes (bacteria) lack a nuclear envelope; eukaryotic cells have a nucleus in which the genetic material is separated from the cytoplasm.
- ❑ Prokaryotic cells are generally smaller and simpler than eukaryotic cells; in addition to the absence of a nucleus, their genomes are less complex and they do not contain cytoplasmic organelles or a cytoskeleton.

Introduction



Table 1.1 Prokaryotic and Eukaryotic Cells

A plasmid is a small, circular DNA molecule that is separate from a cell's chromosomal DNA. Plasmids are most commonly found in bacteria, but can also be found in some archaea and eukaryotic organisms.

Characteristic	Prokaryote	Eukaryote
Nucleus	Absent	Present
Diameter of a typical cell	$\approx 1\mu\text{m}$	$10\text{--}100\mu\text{m}$
Cytoskeleton	Absent	Present
Cytoplasmic organelles	Absent	Present
<u>DNA</u> content (base pairs)	1×10^6 to 5×10^6	1.5×10^7 to 5×10^9
Chromosomes	Single circular <u>DNA</u> molecule	Multiple linear <u>DNA</u> molecules

<https://www.youtube.com/watch?v=zZtcMBTQaS4>
<https://www.youtube.com/watch?v=Pxujitlv8wc>
<https://www.youtube.com/watch?v=q9fthbK08Uc>

Introduction



- ❑ **REPRODUCTION:** Asexual (binary fission) vs Sexual Reproduction
- ❑ **RIBOSOMES** (central dogma of molecular biology)
- ❑ **CHROMOSOMES** (circular vs linear; single vs multiple)
- ❑ **Extrachromosomal material** (plasmid)
- ❑ **MITOCHONDRIA vs CHLOROPLASTS**
- ❑ **EXAMPLES** of Prokaryotes and Eukaryotes
- ❑ **EXAMPLES** of Unicellular Eukaryotes

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<https://www.youtube.com/watch?v=q9fthbK08Uc>

Endosymbiotic theory

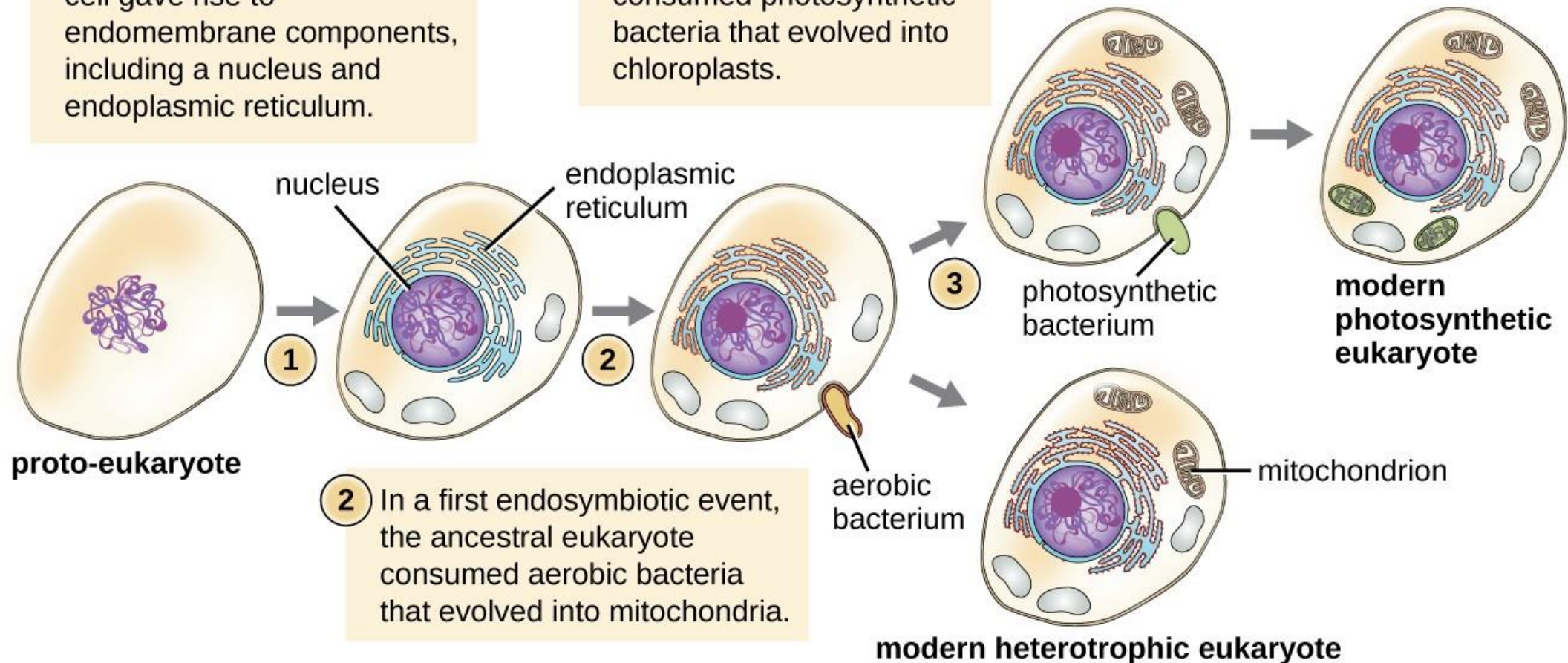
The Endosymbiotic Theory

1 Infoldings in the plasma membrane of an ancestral cell gave rise to endomembrane components, including a nucleus and endoplasmic reticulum.

3 In a second endosymbiotic event, the early eukaryote consumed photosynthetic bacteria that evolved into chloroplasts.

2 In a first endosymbiotic event, the ancestral eukaryote consumed aerobic bacteria that evolved into mitochondria.

Phagocytosis is a process that involves a cell engulfing a large particle using its plasma membrane. This process creates an internal compartment called the phagosome.



Evidence of Endosymbiotic theory



- ❑ Presence of circular DNA (similar genomes to prokaryotes)
- ❑ Presence of 70S ribosomes
- ❑ Can self-replicate (through binary fission)
- ❑ Size (around 10 μm)
- ❑ Have two membranes
- ❑ **Porins** and **cardiolipin** are only found in bacteria, mitochondria and chloroplasts
- ❑ Location of Electron Transport Chain

Porins are proteins that are abundant in the mitochondrial outer membrane. They facilitate the transfer of lipids between the outer and inner membrane, and promote a backup pathway for cardiolipin synthesis in mitochondria.

Cardiolipins are important for regulating mitochondrial proteins, such as carrier proteins, electron transport complexes, and phosphate kinases. They are also essential for organizing mitochondrial structures, such as cristae and contact sites.

Important Links (Endosymbiotic theory)



- ❑ <https://www.youtube.com/watch?v=8oSqXAwLsZc&t=158s>
- ❑ <https://www.youtube.com/watch?v=eCWwKrJyKxA>
- ❑ <https://www.youtube.com/watch?v=FGnS-Xk0ZqU>
- ❑ <https://www.youtube.com/watch?v=bBjD4A7R2xU>
- ❑ <https://www.youtube.com/watch?v=4LhBZ2H5SwM>
- ❑ <https://www.youtube.com/watch?v=Fdh62URT5xE>

Thank you for your attention